



11. The third doctrine of Schwann and Schleiden was disproved by
 - A) Antony van Leeuwenhoek
 - B) Rudolf Virchow
 - C) Ervin Chargaff
 - D) Francois Raspeil
12. Lysosomes are known as "suicidal bags" because of:
 - A) Presence of food vacuole
 - B) Catalytic activity
 - C) Hydrolytic activity
 - D) Parasitic activity
13. Carrier proteins are highly specific for:
 - A) Transported solutes
 - B) Transported solvents
 - C) Glucose transporter
 - D) Un-transported solvents
14. Which of the following is NOT a product of microbial fermentation?
 - A) Antibiotic
 - B) Paper
 - C) Alcohol
 - D) Plastics
15. A cell containing 40 chromatids at the beginning of meiosis would, at its completion, produce cells containing how many chromosomes?
 - A) 20 chromosomes
 - B) 10 chromosomes
 - C) 80 chromosomes
 - D) 40 chromosomes
16. Cell-cell junction which allows the passage of small water-soluble ions and molecules & electrical signals is -----
 - A) Gap junctions
 - B) Tight junctions
 - C) Desmosomes
 - D) Hemidesmosomes
17. Most abundant glycoprotein (about half of the total protein in the body) is -----
 - A) Collagen
 - B) Elastin
 - C) Fibrinogen
 - D) Integrin
18. Which of the following is NOT a characteristic of stem cells?
 - A) Limited proliferative ability
 - B) Self-renewal
 - C) Multipotency or Pluripotency
 - D) Differentiation into specialized cell types
19. Which of the following best describes cellular engineering?
 - A) Modifying cellular structures for industrial applications
 - B) Using cells to engineer tissues and organs
 - C) Altering cellular functions through genetic manipulation
 - D) Studying the morphology of cells under a microscope
20. What type of metabolism is predominantly utilized by microbial cells during fermentation?
 - A) Aerobic
 - B) Anaerobic
 - C) Chemotrophic
 - D) Heterotrophy

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) List out the differences between prokaryotes and eukaryotes.	8	L*2	1	1,12
	b) Compare and contrast optical microscopy and electron microscopy in terms of resolution, magnification, and applications in cell biology.	8	L2	2	1,12
2.	a) Discuss the principles and applications of live cell imaging techniques, using confocal microscopy.	8	L2	2	1,12
	b) Give a detailed account on the stages of mitosis. Add a note on its significance	8	L2	1	1,12
3.	a) Define cell cycle regulation. Give a detailed account on eukaryotic cell cycle regulation. Add a note on its significance.	8	L2	2	1,12
	b) Discuss the principle and applications of electron microscopy with the help of appropriate illustrations.	8	L1	1	1,12
Unit – II					
4.	a) Describe the different types of cytoskeletal elements and their significance in maintaining cell structure and function.	8	L2	3	1,12
	b) Discuss the significance of cell junctions in multicellular organisms, focusing on their role in cell adhesion, communication, and tissue integrity.	8	L2	3	1,12

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|-------|---|-----|----|---|------|
| 5. a) | Discuss the general principles of cell communication, including the roles of signaling molecules, receptors, and intracellular signaling pathways in transmitting signals between cells. | 8 | L2 | 4 | 1,12 |
| b) | Write note on: i) Tight junctions, ii) Desmosomes | 4+4 | L2 | 3 | 1,12 |
| 6. a) | Explain how cells sense the presence of other cells and their environment, including the role of cell surface receptors, cell adhesion molecules, and signaling pathways in mediating cell-cell interactions. | 8 | L2 | 4 | 1,12 |
| b) | Describe: i) Autocrine, ii) Endocrine, signaling, and their respective mechanisms of action. | 4+4 | L2 | 4 | 1,12 |

Unit – III

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|-------|---|---|----|---|------|
| 7. a) | With suitable examples, describe how can microbial cells be utilized in medical diagnosis? | 8 | L2 | 5 | 1,12 |
| b) | Discuss the importance of microbial cells in environmental monitoring. | 8 | L2 | 5 | 1,12 |
| 8. a) | Explain the significance of cell lines derived from animal cells in biomedical research. | 8 | L2 | 5 | 1,12 |
| b) | Explain the concept of cellular engineering and its applications in biotechnology and medicine. | 8 | L2 | 5 | 1,12 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
